

OSSP (Operating Systems And Systems Programming)

Course code: 25CS2104E

Section: 3

Project: University Laboratory Management Server

Project Dashboard Screenshots:

**LABCORE**
University Laboratory Management System

SERVER ONLINE

Dashboard

Request Resource

Student Status

Allocated Resources

Server Response

LABORATORY CONTROL CENTER

Laboratory Dashboard

Monitor laboratory resources and server availability from one centralized interface.

Refresh

System Status

All laboratory resources are being monitored normally.

LIVE RESOURCE STATUS

Resource Availability

LIVE

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LIVE RESOURCE STATUS

Resource Availability

LIVE

COMPUTER

10

Available resources

PRINTER

5

Available resources

FILE SERVER

5

Available resources

PROJECTOR

5

Available resources

SCANNER

5

Available resources

HEADSET

5

Available resources

QUICK ACCESS

Laboratory Operations

Request Resource

Allocate a laboratory resource

Student Status

Check current allocation

Allocations

View active resources

Server Response

View latest server message

LABCORE University Laboratory Management System

Operating Systems Project

**RESOURCE ALLOCATION**

Request Laboratory Resource

Submit a request to allocate a laboratory resource.

New Resource Request[THREAD + SEMAPHORE](#)

The request will be processed by the laboratory management server.

Student ID**Resource****Usage Time**

☐ Resource will be automatically released after the requested duration.

[Request Resource →](#)**STUDENT SERVICES**

Student Resource Status

Check the resources currently allocated to a student.

Search Student Allocation[SERVER QUERY](#)

Enter the student ID to retrieve current laboratory usage information.

[View My Status](#)**Protected Query**

Student allocation information is retrieved directly from the C server.

**Usage Tracking**

Requested, used and remaining resource time can be monitored.



ACTIVE ALLOCATIONS

Allocated Resources

View resources currently assigned to students.

↻ Refresh

Current Allocations

Live allocation information from the laboratory server.

LIVE DATA

```
=====
CURRENT RESOURCE ALLOCATIONS
=====
No resources are currently allocated.
=====
```



SERVER COMMUNICATION

Server Response

View the latest response received from the C laboratory server.

Latest Server Message

Communication between the web interface and C server.

C SERVER

```
Waiting for a request...
```